Reviving an Excel 2000 Easter Egg

If you have not heard of it, Dev Hunter is an Easter egg in Microsoft Excel 2000. It can be triggered by navigating to WC2000 cell and then pressing the Office Logo[1]. It features a 2D shooting game in which you control the car at the center of the screen and shoot any cars in front of you[2].

The Easter egg is called Dev Hunter because it prints the names of the developers on the road and sets the plot that you are "hunting" for them. The game itself is simple, but it was one of the few pastimes for the dull computer classes that I took during high school.

Like all adults, I grew nostalgic for the good things I had when I was young, and I tried to play the game again on modern hardware. This article documents the exploration process.

To start with, I had to use a Windows XP virtual machine to run Excel 2000. But once I followed all the necessary steps, Dev Hunter popped up!

I played the game for a bit and find it no longer very interesting – maybe I only found it interesting because I had little else to do in the computer lab? Anyway, I realized that I am a reverse engineer and should look at the code that implements this Easter egg game.

Remember the first step to launch Dev Hunter is to export the spreadsheet to a webpage – I inspected it and found that it references the CLSID 0002E510-0000-0000-C000-000000000046 which points to MSOWC.DLL[3] in the Office installation directory.

I then loaded this DLL into Binary Ninja and searched for relevant strings. Luckily, the search for "Dev Hunter" immediately sent me to the function 0x3c7dc79b (func1), which contains the game logic. Its parent function, i.e., 0x3c7dc946 (func2), contains the PeekMessageA - TranslateMessage - DispatchMessageA loop that is common for Windows GUI applications.

Upon further investigation, I found that the parent function of func2, i.e., 0x3c7ee86a (func3), contains the logic to decide if the Easter egg should be triggered. In the following code snippet, we can see the row is checked against 0x7cf (1999 in decimal). Similarly, the value 0x2bd represents column WC:

```
cmp dword [ebp-0x8 {column}], 0x2bd
jne 0x3c7ee97b

mov ebx, 0x7cf
cmp dword [ebp-0x14 {row}], ebx
jne 0x3c7ee97b
```

At this point, I could be happy and finish my exploration, but my curiosity drove me further. I wanted to create a "loader" for the game so that it can be run on its own and does not rely on the bulk of Excel installation.

I noticed that func2 is pretty self-contained and can potentially run on its own. It follows thiscall convention. The ecx is a large buffer and probably a C++ class. It is huge, but luckily, the game does not depend on any prior arrangment of its content. So I simply allocate a 0x100000 size buffer, zero it and handle it over via ecx.

There are three parameters on the stack. The first one is a handle to the MSOWC.DLL and the second must be 0x0. The third one is relevant to the colors of the cars but not fully understood. The value 0x7fa87860 used by func3 works well enough.

I wrote the following ASM code[4] that loads the MSOWC.DLL, prepares the parameters, and calls func2:

```
; PAGE_READWRITE
push 0x4
; MEM_COMMIT
push 0x1000
; size
push 0x100000
push 0
call [VA(VirtualAlloc)]
mov edx, eax

; zero the allocated page
mov edi, eax
xor eax, eax
mov ecx, 0x40000
stosd

mov ecx, edx

push VA(Dll)
call [VA(LoadLibraryA)]

push 0x7fa87860
push 0
push eax

; calculate address of func2 and call it
add eax, 0x10c946
call eax
```

I compiled the above code to loader.exe. I also had to make a few patches to func1 and leverage the DDrawCompat[5] project to make it run properly. But it works!

P.S.: The above was achieved back in November 2019. While preparing this article, I noticed that I have not yet figured out how developer names are stored in the DLL. This time I looked closer at the binary and found the string is XOR-ed with byte 0x52. The relevant code is in the loop starting at 0x3c7df36d. The only twist here is that the encryption operates in CBC mode. In case you are curious, please check out the decryption script and result at https://pastebin.com/81T9VzuB.

[1] https://www.vbforums.com/showthread.php?384057-Excel-2000-Secret-Car-Game
[2] https://www.youtube.com/watch?v=B2jlbsmL2fQ
[3] sha1: 3b42043ab53b767cd75a681823138e8a7110dd7a
[4] https://github.com/jeffli678/excel2000-devhunter-loader/blob/master/loader.asm
[5] https://github.com/narzoul/DDrawCompat